

Problem:

Determine whether a day is suitable for playing tennis based on attributes:
Outlook, Temperature, Humidity, Wind.



Q3. FIND-S ALGORITHM - CONCEPT MAP

Goal: Find the most specific hypothesis that is consistent with all positive examples.

FIND-S IDEA

Start with the most specific hypothesis (all $\emptyset$). When a positive example is encountered, generalize the hypothesis minimally to cover that example. Ignore all negative examples.



DATASET

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes ✓
4	Rainy	Mild	High	Weak	Yes ✓
5	Rainy	Cool	Normal	Weak	Yes ✓
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes ✓
8	Sunny	Mild	High	Weak	No

Positive Examples
(Use only these)

FIND-S ALGORITHM (FLOW)

Initialize hypothesis h to the most specific hypothesis (all $\emptyset$)

For each training example x

If x is a positive example

For each attribute, generalize h minimally to be consistent with x

Return the final hypothesis h

SYMBOLS USED

$\emptyset$: Most specific value (no information)

? : Any value (most general value)

h : Hypothesis

We ignore all negative examples.

STEP-BY-STEP HYPOTHESIS UPDATES (ONLY POSITIVE EXAMPLES)

Step	Training Example (Positive Only)	Outlook	Temperature	Humidity	Wind	Hypothesis h (after update)
0 (Initial)	-	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\{\emptyset, \emptyset, \emptyset, \emptyset\}$
1	E3 (Overcast, Hot, High, Weak)	Overcast	Hot	High	Weak	$\{\text{Overcast, Hot, High, Weak}\}$
2	E4 (Rainy, Mild, High, Weak)	?	?	High	Weak	$\{?, ?, \text{High, Weak}\}$
3	E5 (Rainy, Cool, Normal, Weak)	?	?	?	Weak	$\{?, ?, ?, \text{Weak}\}$
4	E7 (Overcast, Mild, Normal, Strong)	?	?	?	?	$\{?, ?, ?, ?\}$

ATTRIBUTE DOMAINS

Outlook = {Sunny, Overcast, Rainy} Humidity = {High, Normal}
Temperature = {Hot, Mild, Cool} Wind = {Weak, Strong}



SUMMARY: Find-S only generalizes when necessary to include positive examples and ignores negative ones completely.

FINAL HYPOTHESIS

$h_{\text{final}} = \{?, ?, ?, ?\}$

This is the maximally specific hypothesis consistent with all positive examples.



KEY TAKEAWAYS

- Find-S returns the most specific hypothesis that covers all positive examples.
- As more positive examples are seen, the hypothesis becomes more general.
- If conflicting positive examples exist, the hypothesis becomes fully general (all ?).



PROBLEM :

Determine whether a person will go for a workout based on attributes:

Weather, Temperature, Time, and Mood.

Apply the Find-S algorithm and determine the maximally specific hypothesis using only positive examples.



Q6. FIND-S ALGORITHM - CONCEPT MAP

GOAL : Find the most specific hypothesis h that is consistent with all positive training examples. Ignore all negative examples.

FIND-S IDEA

- Start with the most specific hypothesis (all $\emptyset$).
- When a positive example is encountered, generalize the hypothesis minimally to cover that example.
- Use only positive examples.



DATASET

Example	Weather	Temperature	Time	Mood	Workout
1	Sunny	Hot	Morning	Tired	No
2	Rainy	Mild	Evening	Happy	Yes ✓
3	Sunny	Mild	Evening	Happy	Yes ✓
4	Cloudy	Cool	Morning	Happy	Yes ✓

Positive Examples
(Use only these)

FIND-S ALGORITHM (FLOW)

Initialize hypothesis h to the most specific hypothesis (all $\emptyset$)

For each training example x

If x is a positive example

For each attribute, generalize h minimally to be consistent with x

Return the final hypothesis h

SYMBOLS USED

$\emptyset$: Most specific value for an attribute (no info)

? : Any value (most general value)

h : Hypothesis



A fit body starts with the right decision! ♥

STEP-BY-STEP HYPOTHESIS UPDATES (ONLY POSITIVE EXAMPLES)

Step	Training Example (Positive Only)	Weather	Temperature	Time	Mood	Hypothesis h (after update)
0 (Initial)	-	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\{\emptyset, \emptyset, \emptyset, \emptyset\}$
1	E2 (Rainy, Mild, Evening, Happy)	Rainy	Mild	Evening	Happy	$\{\text{Rainy, Mild, Evening, Happy}\}$
2	E3 (Sunny, Mild, Evening, Happy)	?	Mild	Evening	Happy	$\{\text{?, Mild, Evening, Happy}\}$
3	E4 (Cloudy, Cool, Morning, Happy)	?	?	?	Happy	$\{\text{?, ?, ?, Happy}\}$

ATTRIBUTE DOMAINS

Weather = { Sunny, Rainy, Cloudy }

Temperature = { Hot, Mild, Cool }

Time = { Morning, Evening }

Mood = { Tired, Happy }



GOALS
☒ Health
☒ Energy
☒ Discipline
☒ Consistency

FINAL HYPOTHESIS

$h_{\text{final}} = \{\text{?, ?, ?, Happy}\}$

This is the maximally specific hypothesis consistent with all positive examples.



Let's Workout!

KEY TAKEAWAYS

- Find-S returns the most specific hypothesis that covers all positive examples.
- As more positive examples are seen, the hypothesis becomes more general.
- If conflicting positive examples exist, the hypothesis can become fully general (all ?).

Small steps today, strong results tomorrow!

PROBLEM :

Determine whether a patient follows a healthy lifestyle based on attributes:

Diet, Exercise, Sleep, and Stress Level.

Apply the Candidate Elimination algorithm to compute the **VERSION SPACE** (S and G boundaries) and show the updates after each training example.

DATASET

Example	Diet	Exercise	Sleep	Stress Level	Healthy Lifestyle
1	Balanced	Regular	Adequate	Low	Yes
2	Balanced	Regular	Poor	Medium	Yes
3	Junk	None	Poor	High	No
4	Balanced	Moderate	Adequate	Low	Yes
5	Junk	Regular	Adequate	High	No

STEP-BY-STEP UPDATES OF S AND G AFTER EACH TRAINING EXAMPLE

Step (Example)	Training Example (Diet, Exercise, Sleep, Stress)	Class (Yes/No)	Specific Boundary S	General Boundary G (Set of Hypotheses)
0 (Initial)	—	—	$(\emptyset, \emptyset, \emptyset, \emptyset)$	$\{?, ?, ?, ?\}$
1 (E1)	(Balanced, Regular, Adequate, Low)	Yes	(Balanced, Regular, Adequate, Low)	$\{?, ?, ?, ?\}$
2 (E2)	(Balanced, Regular, Poor, Medium)	Yes	(Balanced, Regular, ?, ?)	$\{?, ?, ?, ?\}$
3 (E3)	(Junk, None, Poor, High)	No	(Balanced, Regular, ?, ?)	$\{(\text{Balanced}, ?, ?, ?),$ $(?, \text{Regular}, ?, ?),$ $(?, ?, ?, \text{Low})\}$
4 (E4)	(Balanced, Moderate, Adequate, Low)	Yes	(Balanced, ?, Adequate, Low)	$\{(\text{Balanced}, ?, ?, ?),$ $(?, ?, ?, \text{Low})\}$
5 (E5)	(Junk, Regular, Adequate, High)	No	(Balanced, ?, Adequate, Low)	$\{(\text{Balanced}, ?, \text{Adequate}, ?),$ $(\text{Balanced}, ?, ?, \text{Low})\}$

FINAL VERSION SPACE

$S = (\text{Balanced}, ?, \text{Adequate}, \text{Low})$

$G = \{(\text{Balanced}, ?, \text{Adequate}, ?),$
 $(\text{Balanced}, ?, ?, \text{Low})\}$

INTERPRETATION

Any hypothesis h such that $S \leq h \leq G$ is consistent with all the training examples.



KEY TAKEAWAYS

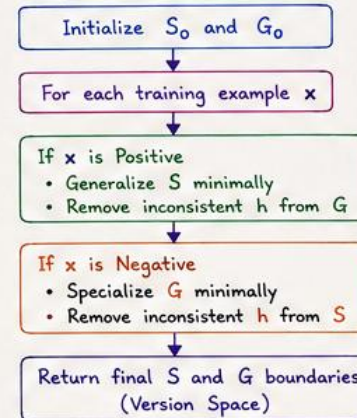
- Candidate Elimination maintains S (most specific) and G (most general) boundaries.
- The version space contains all hypotheses between S and G .
- Positive examples generalize S ; negative examples specialize G .
- Final version space represents all consistent hypotheses.

Q9. CANDIDATE ELIMINATION ALGORITHM - CONCEPT MAP



GOAL : Compute the version space by maintaining the Specific boundary (S) and the General boundary (G) after each training example.

ALGORITHM FLOW



CANDIDATE ELIMINATION - IDEA

- Start with S as the most specific hypothesis and G as the most general hypothesis.
- For each positive example: generalize S and remove inconsistent hypotheses from G .
- For each negative example: specialize G and remove inconsistent hypotheses from S .
- The version space is all hypotheses between S and G .



Healthy Lifestyle!

SYMBOLS USED

- $\emptyset$: Most specific value (no information)
- $?$: Any value (most general value)
- S : Specific boundary (lower bound)
- G : General boundary (upper bound)
- h : Hypothesis



Small changes,
big results!



← Specialize G to exclude inconsistent hypotheses

← Generalize S minimally

← Specialize G again



Live Healthy,
Live Happy!



PROBLEM :

Explain the architecture of a typical machine learning pipeline with a neat diagram.
Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, deployment and monitoring.

Q13. ARCHITECTURE OF A TYPICAL MACHINE LEARNING PIPELINE - CONCEPT MAP



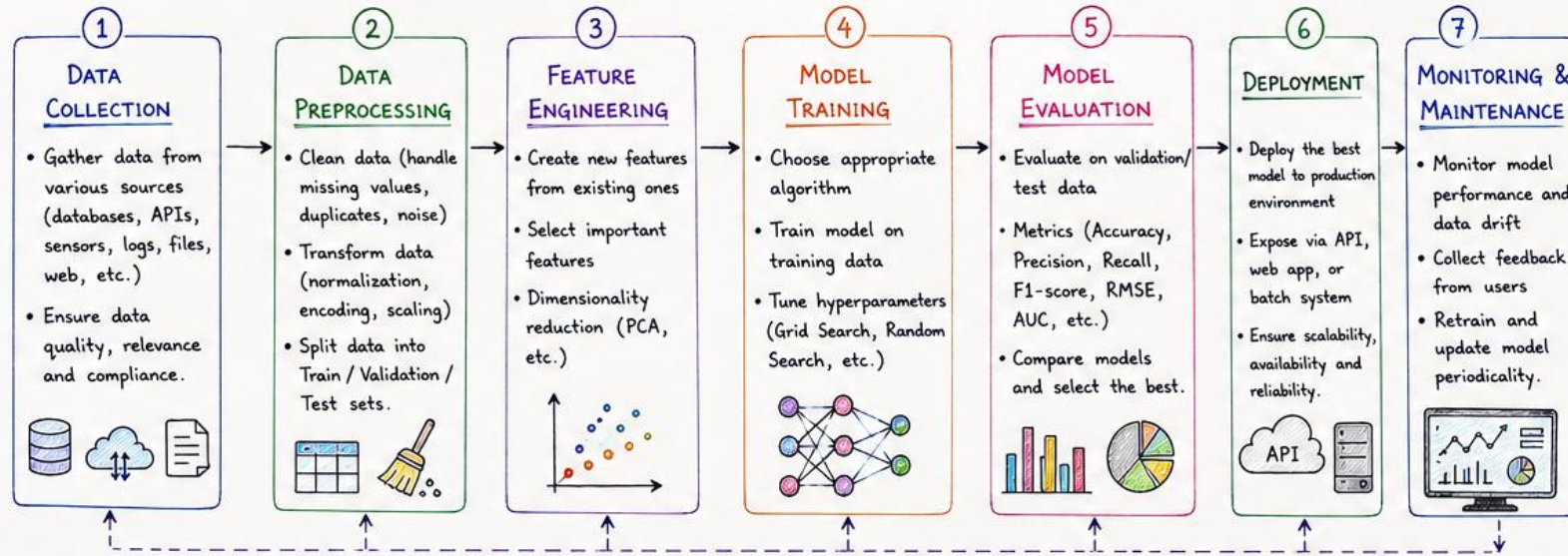
GOAL : Understand the architecture of a typical Machine Learning pipeline and the role of each stage from data collection to deployment and monitoring.

KEY IDEA :

A machine learning pipeline is a sequence of interconnected stages that converts raw data into a deployed model and continuously improves it through feedback.



ML PIPELINE ARCHITECTURE



DESIGN PRINCIPLES

- Modularity
- Automation
- Reproducibility
- Security & Privacy
- Reusability
- Scalability
- Monitoring

OUTCOME

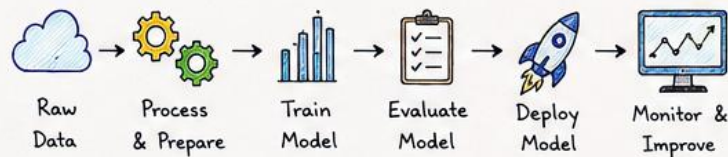
Reliable, accurate and continuously improving ML system in real-world.



FEEDBACK LOOP (CONTINUOUS LEARNING)

Insights from monitoring and feedback are used to collect new data, improve features and retrain the model for better performance.

PIPELINE FLOW (END-TO-END)



TOOLS / TECHNOLOGIES (EXAMPLES)

- Data Collection : Kafka, APIs, DBs
- Preprocessing & ETL : Pandas, Spark
- Feature Engineering : Scikit-learn, Feature Tools
- Training & Evaluation : Scikit-learn, XGBoost, TensorFlow
- Deployment : Flask, FastAPI, Docker, Kubernetes
- Monitoring : Prometheus, Grafana, Evidently AI

SUMMARY

A typical ML pipeline converts raw data into valuable predictions through a series of well-defined stages. Each stage plays a crucial role in building accurate, reliable and scalable machine learning systems.



PROBLEM :

Determine whether a customer will buy a product based on attributes: Age, Income, Student and Credit Rating.

Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis.



Q15. FIND-S ALGORITHM - CONCEPT MAP

GOAL : Find the most specific hypothesis h that is consistent with all positive training examples. Ignore all negative examples.

FIND-S IDEA

- Start with the most specific hypothesis (all $\emptyset$).
- When a positive example is encountered, generalize the hypothesis minimally to cover that example.
- Ignore negative examples completely.
- Final hypothesis is the maximally specific hypothesis that covers all positive examples.



DATASET

Example	Age	Income	Student	Credit Rating	Buys Product
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle	High	No	Fair	Yes ✓
4	Senior	Medium	No	Fair	Yes ✓

Positive Examples
(Use only these)

STEP-BY-STEP HYPOTHESIS UPDATES (ONLY POSITIVE EXAMPLES)

Step	Training Example (Positive Only)	Age	Income	Student	Credit Rating	Hypothesis h (after update)
0 (Initial)	-	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\{\emptyset, \emptyset, \emptyset, \emptyset\}$
1	E3 (Middle, High, No, Fair)	Middle	High	No	Fair	$\{\text{Middle, High, No, Fair}\}$
2	E4 (Senior, Medium, No, Fair)	?	?	No	Fair	$\{?, ?, \text{No, Fair}\}$

ATTRIBUTE DOMAINS

Age = { Young, Middle, Senior } Income = { High, Medium, Low }
Student = { Yes, No } Credit Rating = { Fair, Excellent }



Good data + Good learning = Smart decisions!



FIND-S ALGORITHM (FLOW)

Initialize hypothesis h to the most specific hypothesis (all $\emptyset$)

For each training example x

If x is a positive example

For each attribute, generalize h minimally to be consistent with x

Return the final hypothesis h

SYMBOLS USED

$\emptyset$: Most specific value (no information)

?: Any value (most general value)

h : Hypothesis

We ignore all negative examples.

FINAL HYPOTHESIS

$h_{\text{final}} = \{?, ?, \text{No, Fair}\}$

This is the maximally specific hypothesis consistent with all positive examples.



SUMMARY

Using the Find-S algorithm and considering only the positive examples, the final hypothesis that predicts a customer will buy a product is:

Age = ?, Income = ?, Student = No, Credit Rating = Fair.

